

Monitoring and Logging Network Traffic (3e)

Network Security, Firewalls, and VPNs, Third Edition - Lab 06

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Time on Task:

20 hours, 30 minutes

Progress:

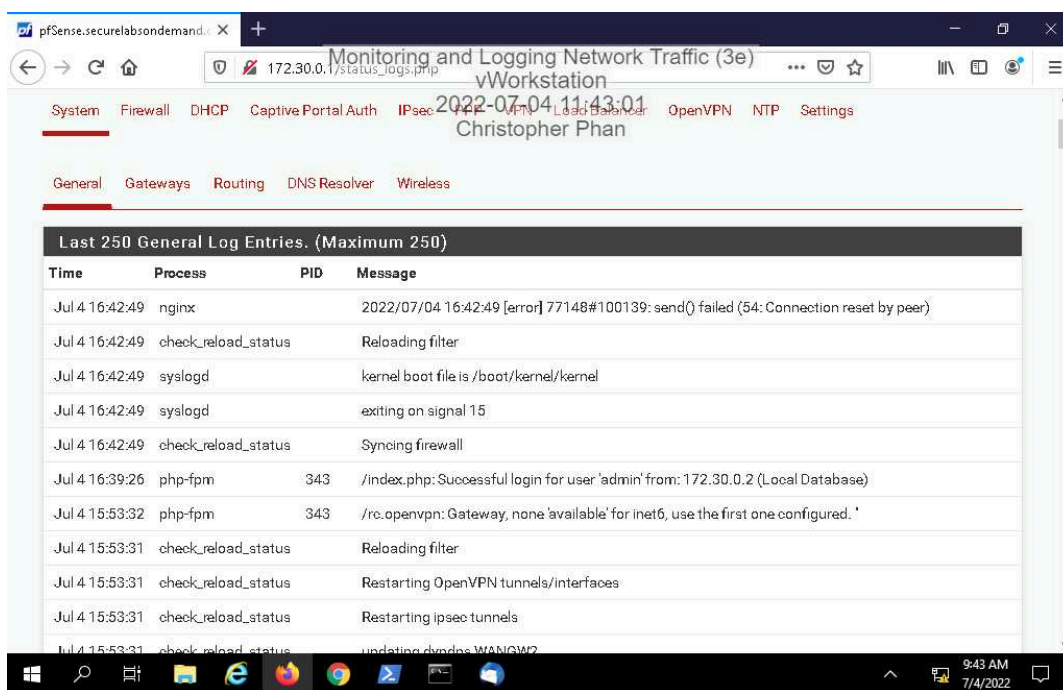
100%

Report Generated: Tuesday, July 5, 2022 at 8:58 PM

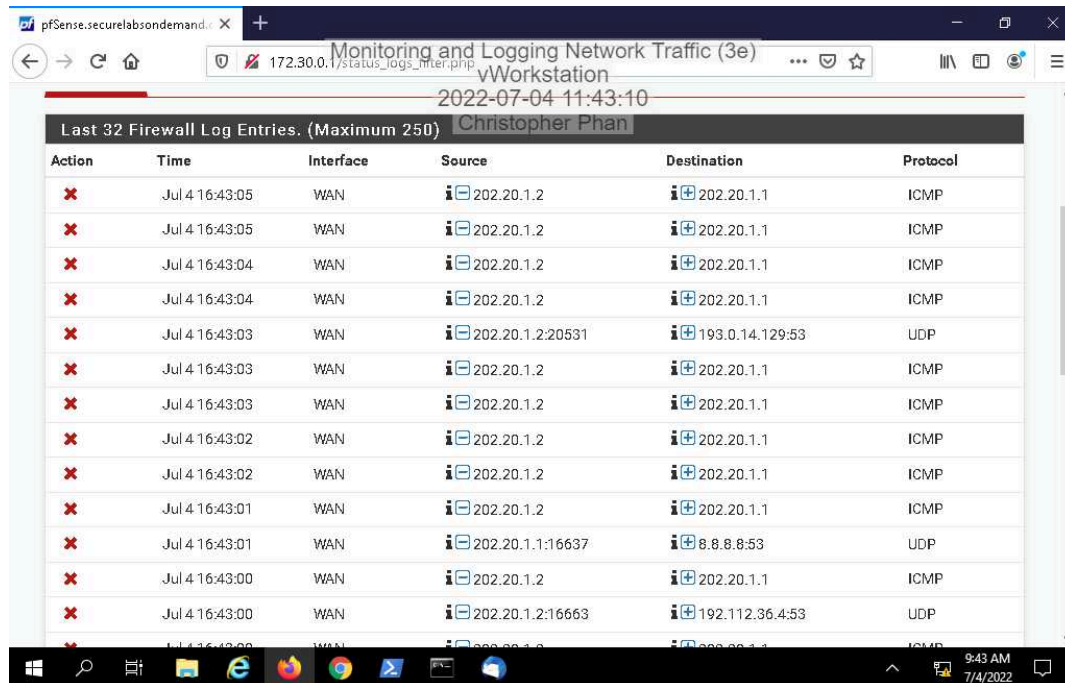
Section 1: Hands-On Demonstration

Part 1: Configure the pfSense Firewall Log

13. Make a screen capture showing the system logs.

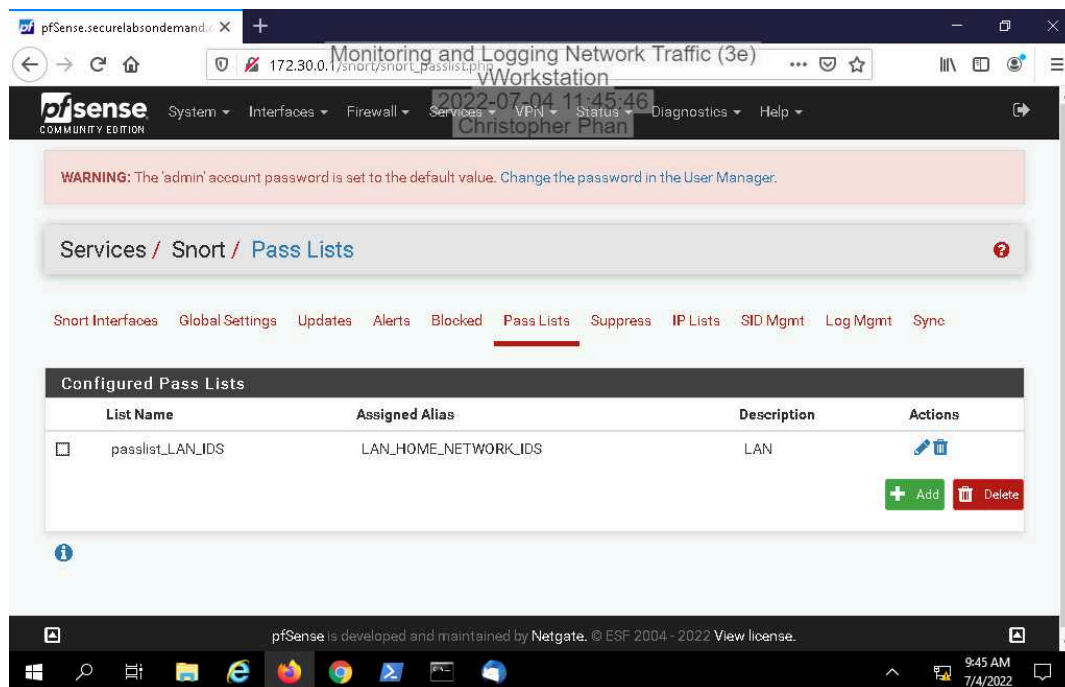


15. Make a screen capture showing the firewall logs.

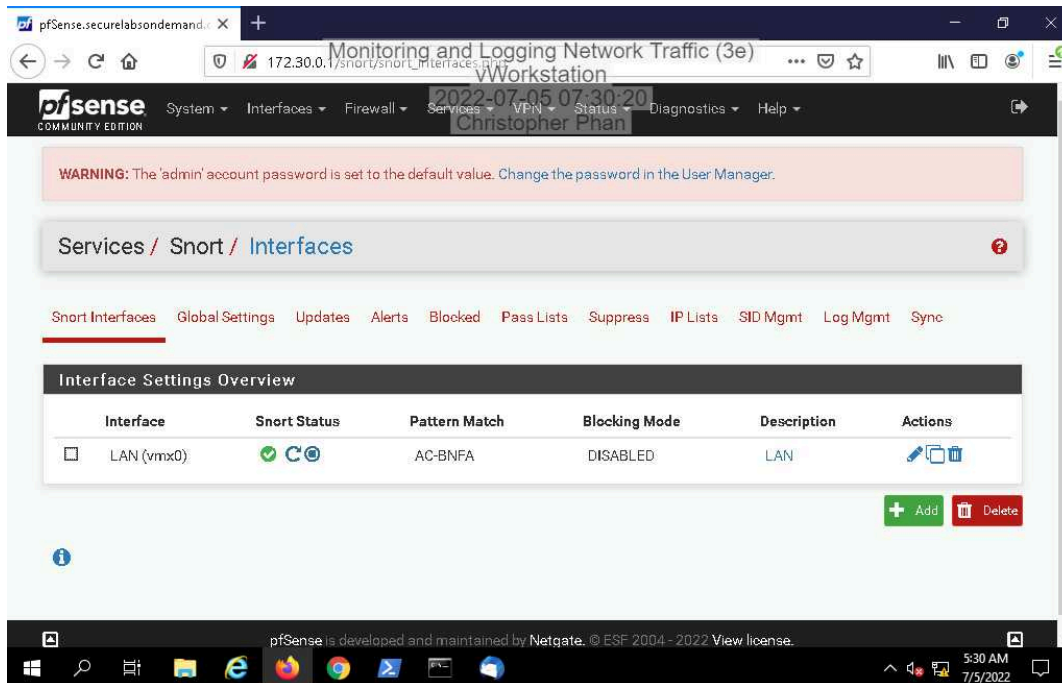


Part 2: Configure a Snort Intrusion Detection System

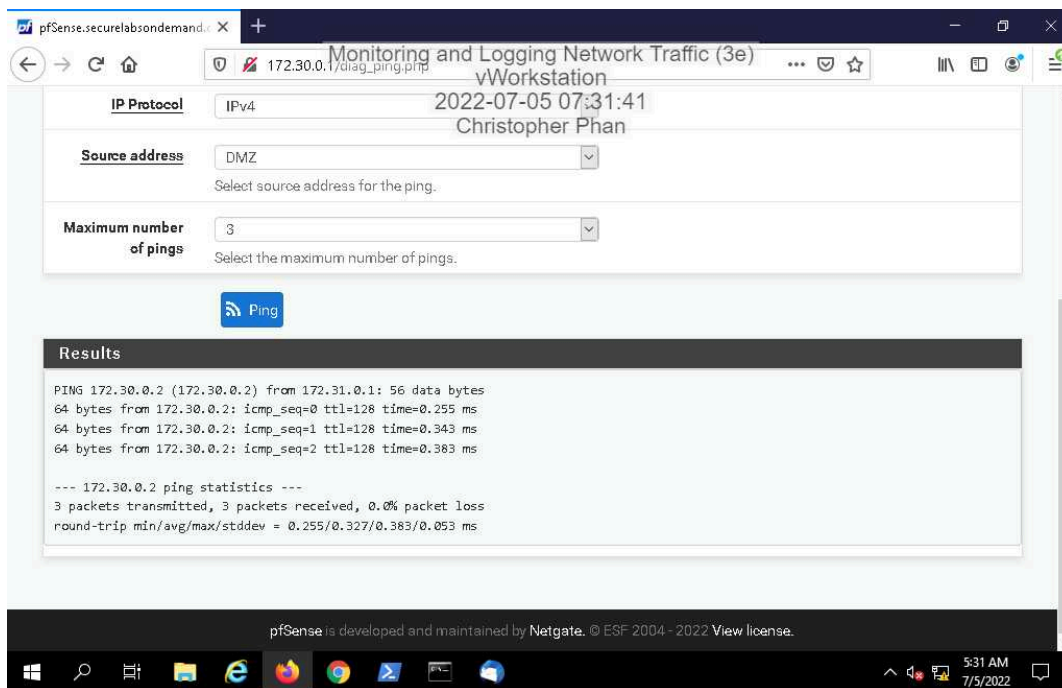
14. Make a screen capture showing the updated Pass Lists page.



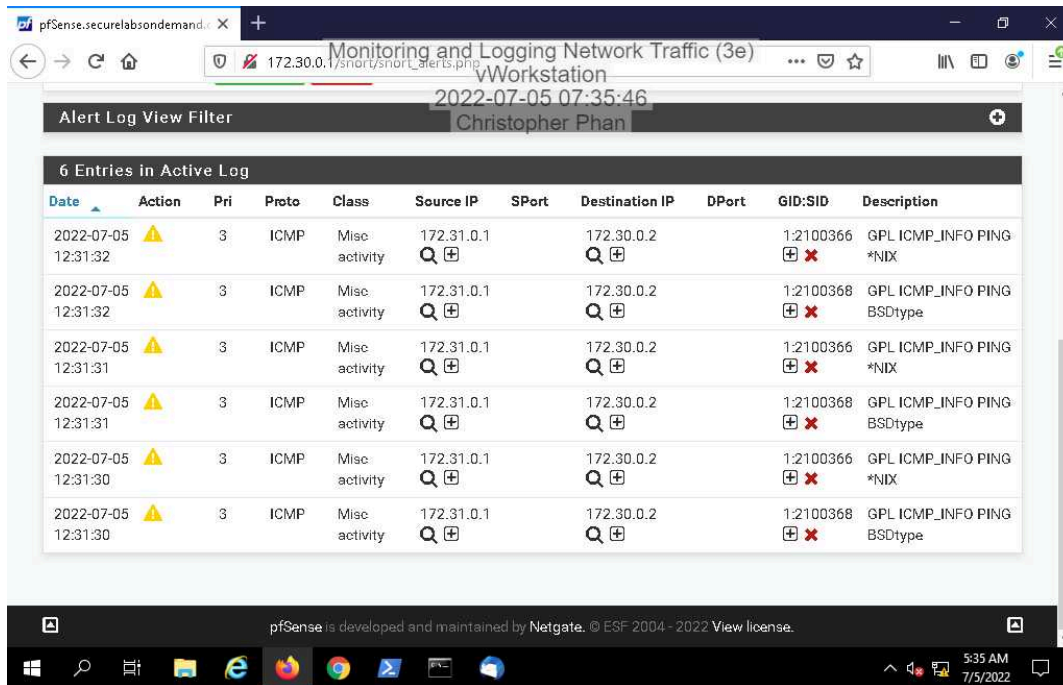
28. Make a screen capture showing the active Snort status on the LAN interface.



33. Make a screen capture showing the successful ping results.

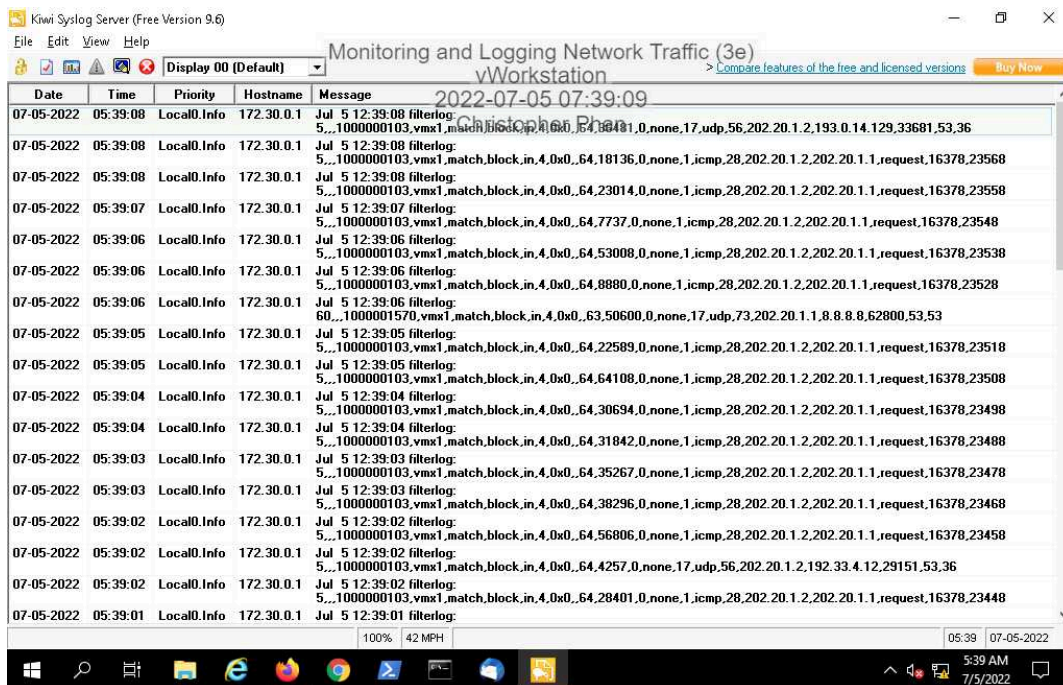


38. Make a screen capture showing the ICMP alerts in the Snort Active Log.



Part 3: Implement Firewall Log Forwarding with Kiwi Syslog Server

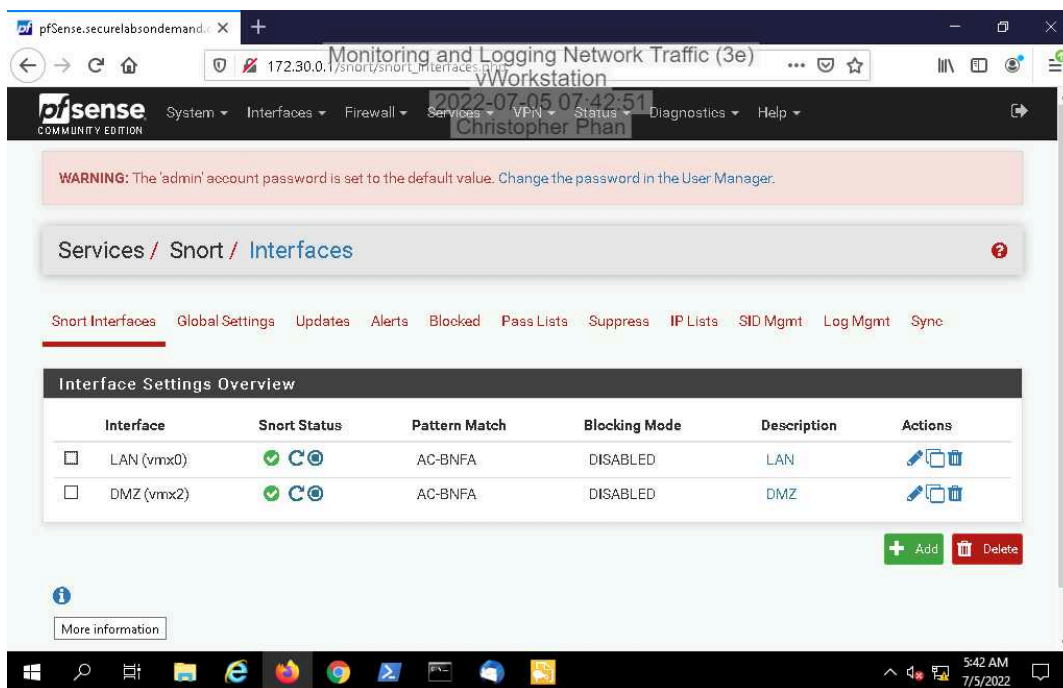
17. Make a screen capture showing the pfSense firewall log events in Kiwi Syslog Server.



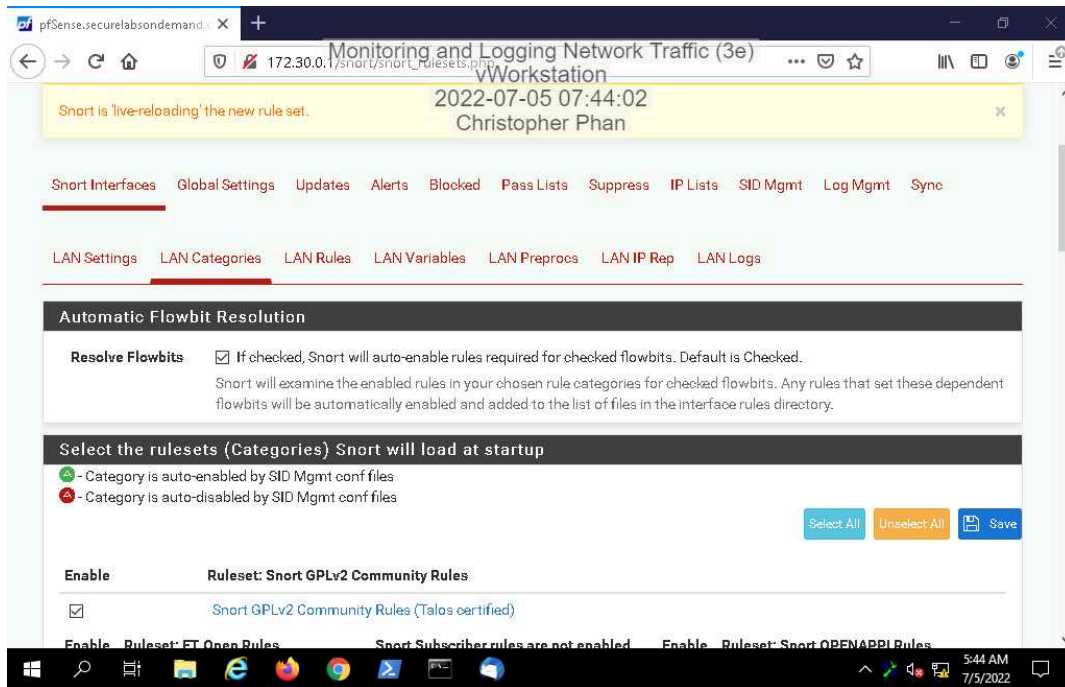
Section 2: Applied Learning

Part 1: Configure Snort Monitoring on the DMZ

17. Make a screen capture showing the active Snort status on the DMZ interface.

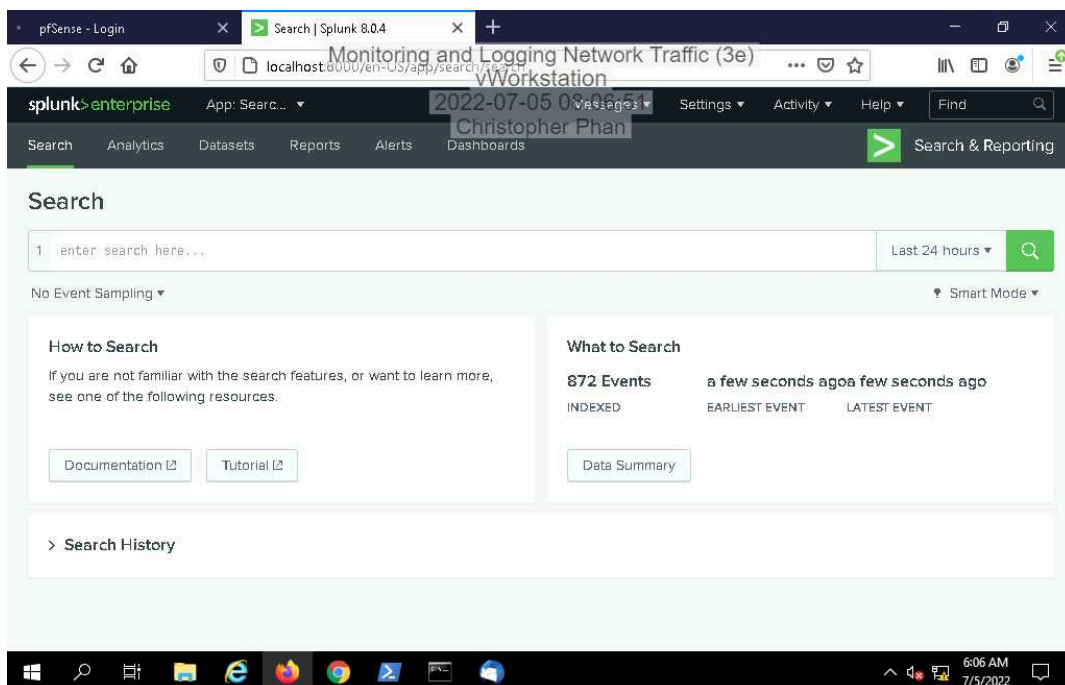


20. Make a screen capture showing the **Snort GPLv2 Community Rules** enabled and "live-reloading" message.



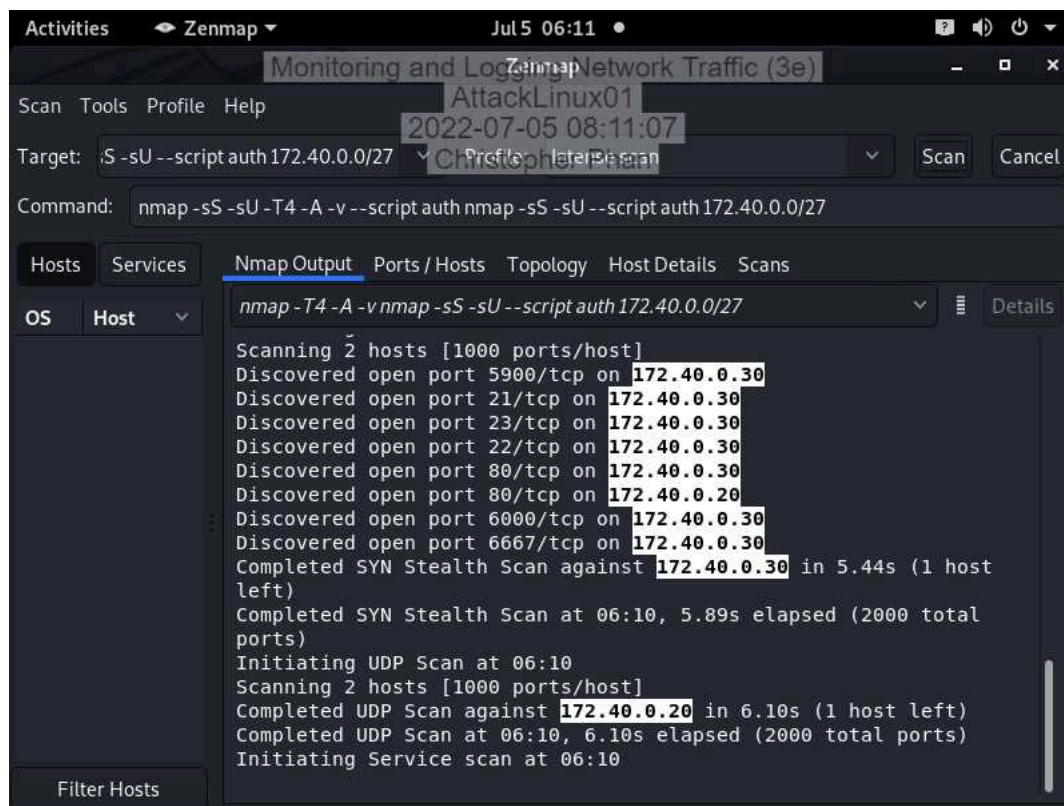
Part 2: Implement Security Information and Event Management with Splunk

13. Make a screen capture showing the indexed events in Splunk.



Part 3: Simulate and Detect a Perimeter Network Attack

6. Make a screen capture showing the Nmap scan report.



9. Make a screen capture showing the search results in Splunk.

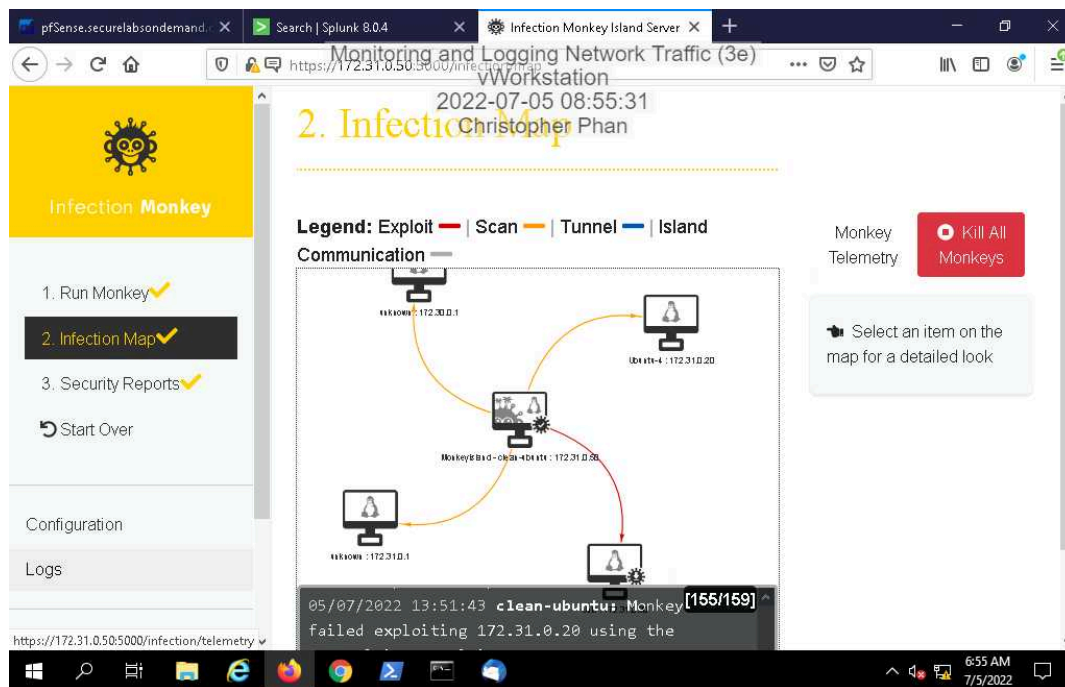
The screenshot shows the Splunk web interface in a browser window. The search bar at the top contains the query: `search snort&sid=1657026697,107`. The search results are displayed in a table with columns for Time and Event. The table shows three events from July 5, 2022, at 06:10:14 AM, 06:06:39 AM, and 06:06:39 AM. The events are related to network traffic analysis, including an invalid content-length or chunk size error and packet processing details. The interface also shows a sidebar with fields like host, source, sourcetype, index, linecount, pid, process, punct, and splunk_server.

Time	Event
7/5/22 6:10:14.000 AM	Jul 5 06:10:14 pfSense.securelabsondemand.com Jul 5 13:10:13 snort[52761]: [120:8:1] (http_inspect) INVALID CONTENT-LENGTH OR CHUNK SIZE [Priority: 3] (TCP) 202.20.1.2:23853 -> 172.31.0.30:80 host = pfSense.securelabsondemand.com source = udp:514 sourcetype = syslog
7/5/22 6:06:39.000 AM	Jul 5 06:06:39 pfSense.securelabsondemand.com Jul 5 13:06:39 snort[52761]: Commencing packet processing (pid=52761) host = pfSense.securelabsondemand.com source = udp:514 sourcetype = syslog
7/5/22 6:06:39.000 AM	Jul 5 06:06:39 pfSense.securelabsondemand.com Jul 5 13:06:39 snort[52761]: Preprocessor Object: SF_FTPTELNET Version 1.2 <Build 13>

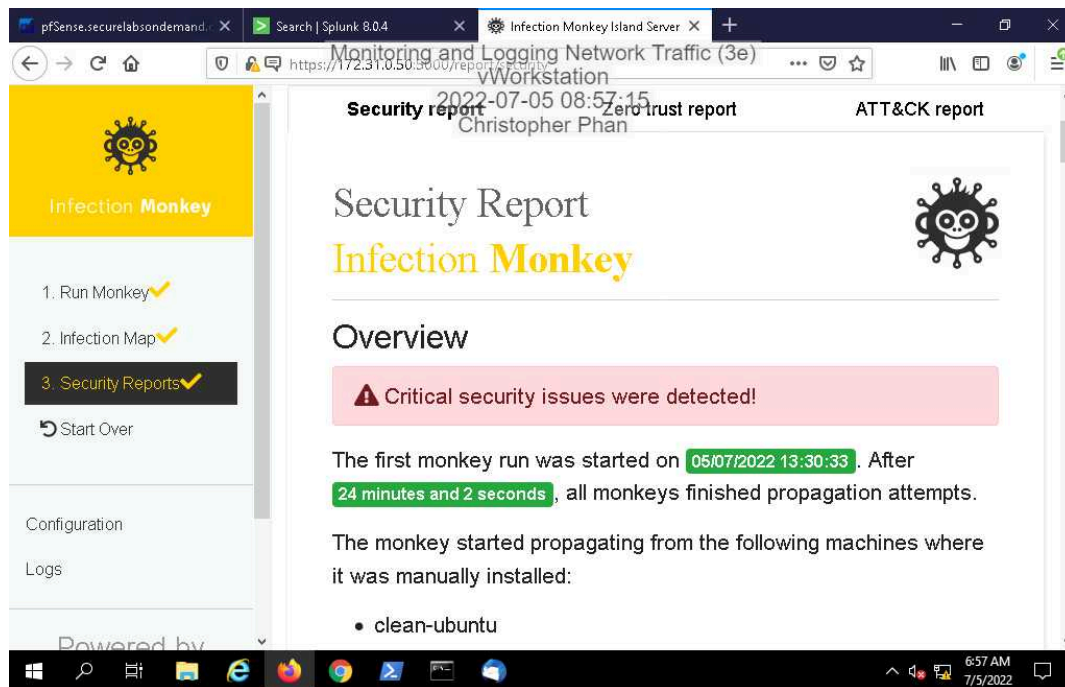
Section 3: Challenge and Analysis

Part 1: Simulate a DMZ Breach with Infection Monkey

Make a screen capture showing the resulting Infection Map.



Make a screen capture showing the resulting Security Report.

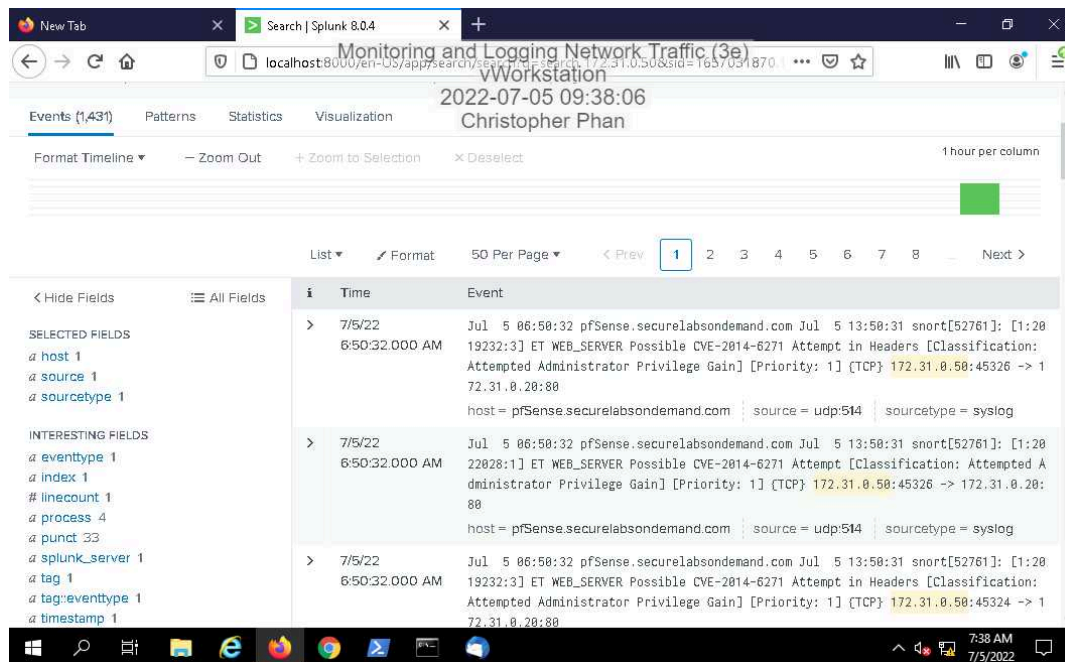


Summarize your DMZ breach simulation results, highlighting what you found to be the greatest concerns from a network monitoring perspective.

The greatest concern for the network is currently one of the Unix system, as it has a unupdated VSFTPD server, which is allowing the network to be exploited.

Part 2: Detect a Simulated DMZ Breach with Snort and Splunk

Make a screen capture showing the **results of your search query for Infection Monkey traffic in Splunk.**



Describe any concerns about the structure of the query result or the data elements it contains. What data fields would you add, remove, or edit to make log analysis more effective?

The problem about the structure of Splunk displaying query results about this type of subject is that it seems clunky, over encumbered with information, hard to read and understand for each entry. What should be done for these data fields is to separate the technical fields and the action shown from the technical fields, making it much more literate for quicker and immediate decisions to be made to stop vulnerabilities.

Write a brief memo to your manager describing Splunk's usefulness in detecting traces of your simulated breach. What configuration changes would you recommend? How would you enhance its functionality?

Splunk is extremely useful in finding and detecting traces of the simulated breaches that occurred through Infection Monkey, as it stores all data that occurred through the system and logs it, easy to be found through key words.